MIL-F-1395A (MU)
3 August 1971
SUPERSEDING
MIL-F-1395 (MU)
20 September 1949

MILITARY SPECIFICATION

FLARE, AIRPORT, M76
PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for and the loading, assembling and packing for one type of flare designated as Flare, Airport, M76.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-P-223	-	Powder, Black
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1234	-	Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1370

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9242383	-	Flare, Airport, M76
8864175	-	Carton, Packing, Ammunition, For Flare, Airport, M76
8864174	-	Box, Packing Ammunition, For Flare, Airport, M76
9200964A	-	Photometer Head
9200965	-	Tube, Sight
9200966	-	Cover
9200967	-	Bracket, Tube
9200968	-	Gasket, Rear
9200969	-	Gasket
9200970	-	Cabinet
9200971A	-	Hood
9200972	-	Bracket
9200973	-	Holder
9200974	-	Spring
9201136C	-	Tunnel
9201268A	-	Light Output
9201390	-	Checkout
9247071	-	Photocell

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 First article inspection.-This specification makes provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2 Material.-The materials shall be in accordance with applicable drawings and specifications.

3.3 Parts and assembly.-The parts and assemblies shall comply with all requirements specified on drawing 9242383 and associated drawings and with all requirements specified in applicable standards and specifications. Packing and packaging shall be in accordance with applicable drawings (see paragraph 5).

3.4 Moisture Content

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3.4.1 Moisture content of pyrotechnic composition.-The moisture content of the pyrotechnic composition at the loading station at the time of loading shall not exceed 0.5 percent when tested as specified in 4.4.1.1.

3.4.2 Moisture content of paper and paper products.-The moisture content of the paper and paper products of the flare assembly at the loading station at the time of loading shall not exceed 6.0 percent when determined as specified in 4.4.1.2.

3.4.3 Moisture content of black powder.-The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.7 percent when tested as specified in MIL-P-223.

3.5 Waterproofness.-The flare shall meet the requirements of 3.6 after being subjected to simulated rain for a period of twenty (20) minutes in accordance with the procedure specified in 4.4.3.

3.6 Functioning.-The flare shall comply with the following requirements when tested as specified in 4.4.4.

3.6.1 The flare igniter shall not function when the pull ring is subjected to a load of two (2.0) pounds or less.

3.6.2 The flare igniter shall function when the pull ring is subjected to a load of not more than six (6.0) pounds.

3.6.3 Ignition of the flare shall be completed in not more than five (5) seconds after the igniter or electric squib has been fired.

3.6.4 The flare burning time shall be not less than five (5) minutes nor greater than seven (7) minutes.

3.7 Workmanship.-All parts and assemblies shall be fabricated, assembled and finished in a thorough, workmanlike manner as described in 3.7.1, 3.7.2 and 3.7.3.

3.7.1 Assembly.-The parts and assemblies shall be complete and shall not have any burrs, sharp or chipped edges, cracks, dents, blisters, warpage or other distortions or signs of damage.

3.7.2 Finishes.-Surface coatings and finishes of parts and assemblies shall be continuous and complete and shall not have scratches or other signs of defective finish or damage that exposes the base metal.

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3.7.3 Cleanliness.-The parts shall be clean and free from dirt, corrosion, grease, rust, corrosion products and foreign matter which would affect appearance or serviceability. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of finding on the moisture content of the black powder, pyrotechnic composition and paper and paper products shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

f. A statement that the components and assemblies were dried and assembled as required by the drawings.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.2c) as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of the following quantities which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract:

a. Twenty-five (25) each of all inert parts and subassemblies.

b. Thirty-two (32) loaded flare assemblies.

c. Twenty (20) illuminant assemblies.

All parts and materials, including packaging and packing components shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for proving ground testing until all other phases of the first article samples have been accepted.

4.2.2 Inspections to be performed

4.2.2.1 Inert parts.-The inert parts and subassemblies will be subjected to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

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4.2.2.2 Loaded assemblies.-The loaded assemblies submitted to a Government approved facility will be subjected by the Government to the functioning test specified in 4.4.4. Prior to functioning sixteen of thirty-two loaded flares submitted will be subjected to the waterproofness test specified in 4.4.3. The illuminant assemblies will be subjected to the test specified in 4.4.2. The test quantities and sequence shall be as follows:

Tests	No. of Assemblies
Illuminant Assemblies	20
Waterproofness	16
Functioning by Squib	8 (Subsequent to Waterproofness) 8 (Controls)
Functioning by Igniter	8 (Subsequent to Waterproofness) 8 (Controls)

4.2.3 Rejection.-If during the performance of any inspection or test specified in 4.2.2 any inert part, sub-assembly or assembly fails to comply with any of the applicable requirements or a defect occurs, the first article sample shall be rejected. The Government reserves the right to discontinue testing upon encountering a basis for rejection.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchaged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall

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comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of flare assemblies shall contain:

a. Primers of one lot interfix number from one manufacturer.

b. Black powder from not more than one lot.

c. Quickmatch from not more than one lot.

d. Pyrotechnic composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Nut (see dwg. 9242416 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection (see 6.4)	Code No. (see 6.3)
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, maximum (max).....	P.D. THD Plug CE (See 6.4.2)	01001

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102.	Minor diameter of thread, max.....	Plain Plug, CE	01002
103.	Depth of thread, minimum (min)....	Min. Effective, Thread Plug Gage, CD (see 6.4.1)	01003

Minor: AQL 1.50 percent

201.	Diameter of nut.....	Ring Gage, CE	01004
202.	Diameter of hole.....	Plug Gage, CE	01005
203.	Protective finish damaged exposing base metal.....	Visual	01006
204.	Evidence of poor workmanship (see 3.7).....	Visual	01007

4.3.2.2 Screw (see dwg. 9242418 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major: AQL 0.65 percent

101.	Pitch diameter of external thread, min.....	P.D. THD Ring Gage, CE	02001
102.	Major diameter of external thread, min.....	SNAP GAGE, CD	02002
103.	Length of thread undercut, max....	Plate Gage, CD	02003
104.	Internal threads missing.....	Visual	02004

Minor: AQL 1.50 percent

201.	Total length.....	Snap Gage, CE	02005
202.	Length from top of head to end of thread.....	Template Gage, CD	02006
203.	Evidence of poor workmanship (see 3.7).....	Visual	02007

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4.3.2.3 Leg (see dwg. 9242415 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Width of leg	Snap Gage, CD	03001
102.	End not spiked	Visual	03002
Minor:	AQL 1.50 percent		
201.	Length of leg	Scale, CE	03003
202.	Dimple missing	Visual	03004
203.	Protective finish damaged exposing base metal	Visual	03005
204.	Evidence of poor workmanship (see 3.7)	Visual	03006

4.3.2.4 Ring (see dwg. 9242398 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Diameter, min	Snap Gage, CE	04001
202.	Thickness, min	Thickness Gage, CE	04002
203.	Evidence of poor workmanship (see 3.7)	Visual	04003

4.3.2.5 Cover, closing (see dwg. 9242397 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter of hole, max	Plug Gage, CE	05001
102.	Diameter of cover, min	Snap Gage, CD	05002

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Minor:	AQL 1.50 percent		
201.	Height of cover.....	Snap Gage, CE	05003
202.	Protective finish damaged exposing base metal.....	Visual	05004
203.	Evidence of poor workmanship (see 3.7).....	Visual	05005

4.3.2.6 Spring (see dwg. 9242394 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Load at test height.....	Spring Tester, CD	06001
102.	Load at assembled height.....	Spring Tester, CD	06002
Minor:	AQL 1.50 percent		
201.	Outside diameter, max.....	Snap Gage, CD	06003
202.	Inside diameter, min.....	Plain Plug Gage, CE	06004
203.	Ends not closed or ground.....	Visual	06005
204.	Evidence of poor workmanship (see 3.7).....	Visual	06006

4.3.2.7 Striker (see dwg. 9242395 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Length of striker point, min...	Optical Comparator, CE	07001
102.	Profile of striker point.....	Optical Comparator, CE	07002
103.	Large outside diameter.....	Ring Gage/ Snap Gage, CD	07003
104.	Diameter of body below flange.....	Ring Gage/ Snap Gage, CD	07004

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105.	Location of fork assembly groove	Max and Min Template Gages, CD	07005
106.	Diameter of fork assembly groove	Max and Min Template Gage, CD	07006
107.	Length to flange	Template Gage, CD	07007

Minor: AQL 1.0 percent

201.	Total length	Snap Gage, CE	07008
202.	Evidence of poor workmanship (see 3.7)	Visual	07009

4.3.2.8 Holder, striker (see dwg. 9242396 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined.

Major: AQL 1.0 percent

101.	Pitch diameter of internal thread, max	THD Ring Gage, CE	08001
102.	Pitch diameter of external thread, min	P.D. THD Ring Gage, CE	08002
103.	Major diameter of external thread, min	Snap Gage, CE	08003
104.	Minor diameter of internal thread, max	Plain Plug Gage, CE	08004
105.	Length of external thread, min	Min Effective thread plug Gage, CD	08005
106.	Diameter of hole, min	Plain Plug Gage, CD	08006
107.	Diameter of striker cavity, min	Special Plug Gage, CD	08007
108.	Depth of striker cavity	Flush Pin Gage, CD	08008
109.	Depth of thread cavity	Min Effective Thread Plug Gage, CD	08009

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Minor:	AQL 1.50 percent		
201.	Length from threaded flange to primer holder end.....	Template Gage, CD	08010
202.	Length from threaded flange to fork assembly end.....	Template, Gage CD	08011
203.	Protective finish damaged exposing base metal.....	Visual	08012
204.	Evidence of poor workmanship (see 3.7).....	Visual	08013

4.3.2.9 Fork (see dwg. 9242390 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length of slot, min.....	Template Gage, CD	09001
102.	Width of slot.....	Plug Gage, CE	09002
103.	Thickness.....	Thickness Gage, CE	09003
Minor:	AQL 1.50 percent		
201.	Opening does not readily admit cotter pin.....	Cotter Pin	09004
202.	Protective finish damaged exposing base metal.....	Visual	09005
203.	Evidence of poor workmanship (see 3.7).....	Visual	09006

4.3.2.10 Ring, pull (see dwg. 9242391 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	10001

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4.3.2.11 Fork Assembly (see dwg. 9242389 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length of fork from bend (min)	Template Gage, CD	11001
102.	Component missing or improperly assembled	Visual	11002
103.	Pull ring does not move freely.	Manual	11003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	11004

4.3.2.12 Cap (see dwg. 9242386 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, max	Thd Plug Gage, CE	12001
102.	Minor diameter of thread, max	Plain Plug CE	12002
103.	Length of perfect thread, min	Min Effective Thd Plug Gage, CD	12003
Minor:	AQL 1.50 percent		
201.	Depth of cavity, min	Flush pin Gage, CD	12004
202.	Knurl missing	Visual	12005
203.	Cap cracked	Visual	12006
204.	Evidence of poor workmanship (see 3.7)	Visual	12007

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4.3.2.13 Hook (see dwg. 9242388 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Loop gap missing	Visual	13001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	13002

4.3.2.14 Cap assembly (see dwg. 9242385 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Cord frayed or worn	Visual	14001
102.	Hook missing	Visual	14002
Minor:	AQL 1.0 percent		
201.	Tape incorrectly applied or missing	Visual	14003
202.	Evidence of poor workmanship (see 3.7)	Visual	14004

4.3.2.15 Holder, primer (see dwg. 9242393 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, min.	THD Ring Gage, CE	15001
102.	Major diameter of thread, min.	Snap Gage, CE	15002
103.	Diameter of hole, max.	Plain Plug Gage, CE	15003

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Minor:	AQL 2.50 percent		
201.	Total length	Snap Gage, CE	15004
202.	Outside diameter	Snap Gage, CE	15005
203.	Length to flange	Template Gage, CD	15006
204.	Protective finish damaged exposing base metal	Visual	15007
205.	Evidence of poor workmanship (see 3.7)	Visual	15008

4.3.2.16 Primer holder assembly (see dwg. 9242392 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Primer missing or damaged	Visual	16001
102.	Cement missing	Visual	16002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	16003

4.3.2.17 Igniter assembly (see dwg. 9242384 covering a detail of 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Any component missing or incorrectly assembled	Visual	17001
102.	Any component loose or not secure	Manual	17002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	17003

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4.3.2.18 Ring, reinforcing (see dwg. 9242412 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.50 percent		
201.	Large diameter	Plug Gage, CE	18001
202.	Diameter of hole	Plug Gage, CE	18002
203.	Height of projection	Template Gage, CD	18003
204.	Diameter inside projected area	Plug Gage, CE	18004
205.	Length	Template Gage, CD	18005
206.	Evidence of poor workmanship (see 3.7)	Visual	18006

4.3.2.19 Priming disc assembly (see dwg. 9242408 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Black powder missing	Visual	19001
102.	Igniting charge composition missing	Visual	19002
103.	Any quickmatch missing or improperly assembled	Visual	19003
Minor:	AQL 1.0 percent		
201.	Staple(s) missing	Visual	19004
202.	Evidence of poor workmanship (see 3.7)	Visual	19005

4.3.2.20 Body assembly (see dwg. 9242405 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Seam overlap, min	Scale	20001

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Minor:	AQL 1.0 percent		
201.	Length, max	Scale	20002
202.	Evidence of poor workmanship (see 3.7)	Visual	20003

4.3.2.21 Cover, base (see dwg. 9242403 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Thickness	Thickness Gage, CD	21001
202.	Inside diameter	Plain Plug Gage, CE	21002
203.	Inside length	Depth Plug Gage, CD	21003
204.	Protective finish damaged exposing base metal	Visual	21004
205.	Evidence of poor workmanship (see 3.7)	Visual	21005

4.3.2.22 Bracket, leg (see dwg. 9242404 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Width of channel(s)	Template Gage, CD	22001
102.	Depth of channel(s)	Template Gage, CD	22002
Minor:	AQL 1.50 percent		
201.	Thickness	Thickness Gage, CE	22003
202.	Length across flats	Template Gage, CD	22004
203.	Evidence of poor workmanship (see 3.7)	Visual	22005

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4.3.2.23 Base cover assembly (see dwg. 9242402 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Any weld missing or inadequate....	Visual-Manual	23001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	23002

4.3.2.24 Illuminant assembly (see dwg. 9242401 covering a detail of dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Designated area not soldered.....	Visual	24001
102.	First fire composition missing....	Visual	24002
103.	Pyrotechnic composition missing...	Visual	24003
104.	Indicated surface not coated with black powder.....	Visual	24004
105.	Any component missing or improperly assembled.....	Visual	24005
106.	Priming disc loose.....	Visual	24006
Minor:	AQL 1.0 percent		
201.	Location of squib opening.....	Gage, Location,CD	24007
202.	Evidence of poor workmanship (see 3.7).....	Visual	24008

4.3.2.25 Flare, surface: Airport, M76 (see dwg. 9242383).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cotter pin missing.....	Visual	25001

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Major:	AQL 0.65 percent		
101.	Any component missing or incorrectly assembled	Visual	25002
102.	Sealing compound missing from designated area	Visual	25003
103.	Instruction label missing	Visual	25004
Minor:	AQL 1.50 percent		
201.	Total length, max	Length Gage, CD	25005
202.	Marking misleading or unidentifiable	Visual	25006
203.	Evidence of poor workmanship (see 3.7)	Visual	25007

4.3.2.26 Carton packing, ammunition, prior to sealing (see dwg. 8864175).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Flare inverted	Visual	26001
102.	Packing components missing	Visual	26002
Minor:	None defined.		

4.3.2.27 Carton, packing, ammunition, after sealing (see dwg. 8864175).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container damaged	Visual	27001
102.	Contents loose	Manual	27002
103.	Bag punctured, cut or torn	Visual	27003
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	27004

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4.3.2.28 Box, packing, ammunition (see dwg. 8864174).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Contents exposed due to damaged box	Visual	28001
102.	Hardware or strapping missing, broken or loose	Visual-Manual	28002
103.	Box broken, board loose or protuding nails	Visual	28003
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	28004
202.	Hardware or strapping improperly assembled	Visual-Manual	28005
203.	Marking misleading or unidentifiable	Visual	28006

4.3.3 Testing

4.3.3.1 Moisture content of pyrotechnic composition and paper and paper products (see Table II).-The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification at least one sample of each material from each eight hours production of flares. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the samples together with all assemblies loaded with the non-conforming material shall be rejected. The contractor shall maintain adequate records to provide identification of the specific explosive mixture and loaded items which are represented by the sample. The tests shall be performed as specified in 4.4.1.1 and 4.4.1.2.

4.3.3.2 Moisture content of black powder (see Table II).- Sample size, test procedure and acceptance criteria shall be as specified in MIL-P-223.

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4.3.3.3 Candlepower (see dwg. 9242401 and Table II).- Twenty (20) illuminant assemblies shall be selected from each lot for test. If two or more assemblies fail to comply with the requirement, the lot shall be rejected. The procedure shall be as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.4 Waterproofness (see 3.5 and Table II).-Twenty (20) flares shall be selected from each lot for this test. The flares subjected to this test shall be retained for subsequent functioning. The test procedure shall be as specified in 4.4.3.

4.3.3.5 Functioning (see Table I).-The flare assemblies shall be observed for any evidence of failure to comply with the requirements, as classified in Table I, when tested as specified in 4.4.4.

4.3.3.5.1 Beginning with the first lot and continuing until three consecutive lots have complied with the applicable requirements specified, eighty (80) flare assemblies shall be selected from each lot for test. The lot shall be rejected if four (4) or more assemblies exhibit any of the defects in Table I.

4.3.3.5.2 After three consecutive lots have complied with the acceptance critera of 4.3.3.5.1, thirty two (32) flare assemblies shall be selected from each lot for test. The lot shall be rejected if two or more assemblies exhibit any of the defects listed in Table I.

4.3.3.6 Embrittlement of spring (see dwg. 9242394 and Table II).-Fifty (50) springs shall be selected from each heat treatment lot for test. If one or more springs fail to comply with the requirement, the heat treatment lot shall be rejected. The test shall be performed as specified in 4.4.5. Springs used for this test shall not be returned to the lot.

4.3.3.7 Heat seal of packing bags (see dwg. 8864175 and Table II).-Thirteen (13) sealed packing bags shall be selected from each lot for test. If any heat seal fails to comply with the requirement specified, the lot shall be rejected. This test shall be performed in accordance with the procedure specified in MIL-P-116.

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4.3.3.8 Check test for possible deterioration of primers (see applicable primer specification).-If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into the flare assemblies exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lots shall be subjected to and must satisfactorily pass the check test specified in the applicable primer specification. The check test shall be performed by the contractor (see 6.7) prior to assembling the primers into the signals.

TABLE I

SUMMARY OF FUNCTIONING DEFECTS

DEFECTS	DEFECT CODE	CLASSIFI-CATION	PARA. REFERENCES REQM'T	TEST MTHD	METHOD OF INSP.
Pin load incorrect	25008	Major	3.6.1 & 3.6.2	4.4.4	Pull tester
Flare fails to ignite	25009	Major	3.6.3	4.4.4	Visual
Ignition delay incorrect	25010	Major	3.6.3	4.4.4	Stop Watch
Burning time incorrect	25011	Major	3.6.4	4.4.4	Stop Watch

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TABLE II

PRODUCTION LOT ACCEPTANCE

TEST	DEFECT CODE	CLASSIFICATION	PARAGRAPH REFERENCES REQUIREMENT	TEST METHOD	METHOD OF INSPECTION
Moisture Test for Pyrotechnic Composition	29001	Major	3.4.1	4.4.1.1	
Moisture Test for Paper and Paper Products	29002	Major	3.4.2	4.4.1.2	
Moisture Test for Black Powder	29003	Major	3.4.3	MIL-P-223	
Test for Candlepower	24009	Major	Dwg. 9242401	4.4.2	Mandatory (M) See 6.4.4.1 9200964A Photometer Head 9200965 Tube, Sight 9200966 Cover 9200967 Bracket, Tube 9200968 Gasket, Rear 9200969 Gasket 9200970 Cabinet 9200971A Hood 9200972 Bracket 9200973 Holder 9200974 Spring 9201136C Tunnel 9201268A Light Output 9201390 Checkout 9247071 Photocell

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TABLE II (Con'd)

TEST	DEFECT CODE	CLASSIFICATION	PARAGRAPH REFERENCES REQUIREMENT	PARAGRAPH REFERENCES TEST METHOD	METHOD OF INSPECTION
Test for Waterproofness	29004	Major	3.5	4.4.3	
Spring Embrittlement Test	06007	Major	Dwg. 9242394	4.4.5	
Test for Heat Seal	29004	Major	Dwg. 8864175	MIL-P-116	

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4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is described under the "Method of Inspection" column in paragraphs 4.3.2 and 4.3.3. Special Inspection Equipment (SIE) designs, if required, are specified by drawing numbers (see 6.4.3). When no SIE design is provided, the contractor shall submit for approval inspection design in accordance with 6.4.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings provided and revised in accordance with the requirements herein may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever isnpection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Pyrotechnic composition.-Determination of the moisture content of pyrotechnic composition shall be performed by transferring five (5) grams of the finely divided sample to a tared moisture dish. Dry the sample in a vacuum desiccator containing sulphuric acid for a minimum of twenty four (24) hours. Calculate the loss in weight of the dish and contents as the percentage of moisture in the sample.

4.4.1.2 Paper and paper products.-The moisture content shall be determined in accordance with Method 102.1 of MIL-STD-1234, except that a ten (10) gram sample of each paper product shall be used and dried for two (2) hours.

4.4.2 Candlepower.-The illuminent assembly shall be supported in a suitable fixture with the ignition end up and ignited. The candlepower shall be determined and observation made for compliance with the applicable requirement.

4.4.3 Waterproofness.-The flares shall be placed in the test chamber in an upright position and subjected to a water spray (simulated rainfall) of four plus or minus one inch (4 $\pm$ 1 inch) per hour for twenty minutes. A rain gage shall be placed within six inches of the flares to be tested, to measure the rate of water fall. The temperature of the water shall be maintained

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between 52°F and 70°F. The water spray shall be directed in a manner which will cause the droplets to impinge directly on the flares. The spray head shall not be nearer than three feet to any flare under test. After testing, the flares shall be inspected for damage. These flares shall be equally divided and retained for subsequent functioning test; one group to be ignited by firing pin igniter, and one group by electrical ignition (see 4.4.4.).

4.4.4 Functioning.-The test flare assemblies shall be mounted by inserting the legs into the base sockets, spiked ends downward, and pressing legs into the ground.

a. One half of the sample flares shall be ignited by firing pin igniter. To function, unscrew knurled cap, remove tape around lanyard, remove lanyard from cap and hook lanyard into the pull ring. After the safety cotter pin is removed, gradually apply a horizontal pull force to the pull ring. Increase the pull and record the value of the force required to function the flare.

b. One half of the sample flares shall be ignited electrically. Remove the tape holding the squib wires around the body at the top end of the flare. The flares shall be fired using a suitable power source supplying a minimum current of 1.0 ampere to the electric squib. Observation shall be made to determine compliance with the applicable requirements.

4.4.4.1 Proving ground tests.-These tests shall be conducted at a Government Proving Ground in accordance with the Aberdeen Proving Ground Acceptance Test Procedure, FL-4.

4.4.4.2 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.5 Embrittlement of spring.-The spring shall be stretched along its longitudinal axis until it no longer returns to its original position. When the spring no longer returns to its original length or profile it shall be considered permanently distorted. Any spring that breaks during the test shall be classed defective. Springs subjected to test shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-Preservation and packaging for Flare, Surface, Airport, M76, shall be in accordance with dwg. 8864175.

5.2 Packing

5.2.1 Level A.-Packing for Flare, Surface, Airport, M76 shall be in accordance with dwg. 8864174.

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5.3 Marking.-Unit and exterior containers shall be marked in accordance with instructions contained in drawings 8864174 and 8864175.

6. NOTES

6.1 Intended Use.-The Flare, Airport, M76, covered by this specification is designed to provide a means of illuminating landing areas during conditions of poor visibility.

6.2 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Quantity required and delivery schedules.
e. Inspection equipment design requirements (see 4.3.4 and 6.4).

6.3 Defect code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers and specification numbers shall be referenced on drawings of equipment designs submitted by the contractor to the Government for approval.

6.4 Inspection Equipment Designs.-The contractor shall design inspection equipment as required under the Method of Inspection column of the applicable paragraphs of Section 4. The contractor shall prepare designs for all inspections for which Special Inspection Equipment (SIE) designs have not been supplied by the Government. If SIE designs are supplied by the Government but are non-mandatory, the contractor may elect to use an approved alternate design (CD) or commercial equipment (CE). The designs shall meet the criteria described in each SIE envelope drawing supplied by the Government. Designs shall be prepared to the Category and From (per MIL-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance.

6.4.1 Contractor designs.-Contractor designs (designated "CD" in the Method of Inspection column of the applicable paragraphs of Section 4) are designs of inspection equipment which the Government has assigned design responsibility to the contractor, or designs of inspection equipment which the contractor has elected to prepare in lieu of non-mandatory SIE designs. Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures.

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6.4.2 Commercial equipment.-Commercial equipment (designated "CE" in the Method of Inspection column of the applicable paragraphs of Section 4) is unmodified inspection equipment which is catalogued and available for purchase by the general public. Commercial equipment which the contractor proposes to use shall be fully described by means of catalog sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared for any special fixtures required to be used with commercial equipment.

6.4.3 Special Inspection Equipment (SIE) designs.-An SIE design is defined herein:

a. Detailed drawings which completely depict all the information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing as defined in MIL-STD-100. This establishes the criteria which a detailed design shall meet.

6.4.4 Classification of SIE Designs

6.4.4.1 Mandatory (designated "M" in the "Method of Inspection " column of the applicable paragraphs of section 4).- The contractor shall use this design to construct equipment for the specified inspection. In the mandatory SIE design is an envelope drawing, the specified criteria shall be met in the detailed design prepared by the contractor. When the method of inspection references mandatory (M) designs, the contractor shall comply with and use these designs only.

6.4.4.2 Non-mandatory.-The contractor may use the SIE design supplied, or may design alternate inspection equipment or propose commercial inspection equipment to facilitate his operations.

6.5 Submission of contractor designs.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review

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will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt. Unless otherwise specified, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND 5, Dover, New Jersey 07801. Unless otherwise specified, all other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract, submission shall be as directed by the contracting officer.

6.6 Submission of test data.-In addition to the normal distribution of records, one copy of the required ballistic test data shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND5 and
SMUPA-DE1
Dover, New Jersey 07801

6.7 Cost of check test.-The contracting officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the flare and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer lots were under the jurisdiction of either contractor.

6.8 Proving Ground Test Summary:

	Sample Size	Test	Requirement
First Article	16	Waterproofness	3.5
	32	Functioning	Table I
Regular Production (First Three Consecutive acceptable lots)	20	Waterproofness	3.5
	80	Functioning	Table I
Remaining	20	Waterproofness	3.5
	32	Functioning	Table I

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1370-A-283

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